

Ravindra Todmal

Ravindra.todmal@gmail.com Cell: +91-98 227 92128

Versatile and goal-oriented professional, targeting **Global Engineering Leadership** roles in **Embedded Systems, Hardware Design** to drive innovative product development, regulatory compliance, and business growth while building high-performing, cross-functional engineering teams.

Executive Summary

- **Visionary engineering leader** with **30+ years** of experience in **driving innovation** and **global product strategies** in **Medical Devices, Automotive, Industrial Automation** and **IOT Technologies**.
- Expert in **full lifecycle product management**, including **NPI, NPD, product sustenance, obsolescence management**, and value engineering, delivering both technical excellence and business growth.
- Skilled in cross-functional team leadership, building and mentoring **high-performing teams of 200+ engineers across geographies**, fostering a culture of innovation, compliance, and operational excellence.
- Displayed expertise in **embedded systems, firmware-controlled electromechanical devices, high-speed PCB design, Hardware-software co-design, system engineering**, and **reliability engineering**, translating complex concepts into commercially viable solutions.
- Recognized for **bridging business and technology** by combining **executive leadership** with hands-on **expertise in embedded systems, electromechanical devices, and high-speed electronics design** to deliver innovative, commercially viable, and fully **regulatory-compliant product solutions**.
- **Strategic driver of digital transformation initiatives**, including **AI-enabled platforms, IoT-enabled devices** and **cloud-connected healthcare solutions**, achieving measurable process efficiency and compliance.
- Recognized for **business impact through innovation** and **value creation**, including successful establishment of regulatory Centers of Excellence, and **delivering multimillion dollar cost savings** through value engineering and lifecycle optimization.
- Proficient in regulatory frameworks and **quality standards (FDA 510(k), ISO 13485, ISO 14971, IEC 60601, IEC 62304, EU-MDR)**, enabling accelerated product approvals and global market entry.

Core Competencies

- Hardware Software Co-design, Real Time Embedded Systems Engineering
- Electromechanical Systems Design, Integration, Validation and Compliance
- New Product Introduction (NPI) & New Product Development (NPD)
- Systems Engineering & V-Model Lifecycle Management
- Product Lifecycle Management (PLM, Sustenance, Obsolescence Management, Reliability Engineering)
- Global Engineering Delivery & Distributed Team Leadership
- Design Center Leadership, P&L & Practice Management
- Value Engineering & Cost Optimization
- High-Speed PCB Design, Signal Integrity
- Regulatory Compliance & Quality Systems (FDA 510(k), ISO 13485, IEC 60601, IEC 62304)

Key Highlights across the Career

- Led end-to-end NPD of a blood apheresis device from concept to manufacturing handover, delivered time-bound product development, thus successfully achieving a predetermined product launch date set during the conceptualization phase.
- Designed and launched wearable medical devices, including a continuous glucose monitoring system (CGMS) that is waterproof, wearable for 15 days, and designed for daily patient use, thereby supporting client product launches.
- Established and scaled hardware and embedded systems engineering teams from scratch at Capgemini and eInfoChips, growing them into high-performing global services units of 200+ engineers.
- Implemented structured value engineering programs across multiple projects, achieved 20-30% cost savings without compromising regulatory compliance or product quality.
- Orchestrated global engineering collaboration and knowledge-sharing practices, thus enhancing cross-site productivity and reduced resource bottlenecks.

Work Experience

Sutherland, Pune India (May 2024 to present)

AVP, Practice Head, Medical Devices mechatronics

- Led the adoption of AI-enabled quality and CAPA assessment platforms that resulted in potential improvement of defect detection and resolution efficiency by 30%.
- Build team of embedded systems, electrical, mechanical and firmware engineers for medical device imaging customers for X-Ray, CT, MRI
- Spearheaded a cross-functional initiative to optimize medical device product design processes, resulting in ISO13485 certification and accelerated time-to-market.

Capgemini, Pune India (Jan 2018 to April 2024)

Director, Medical Devices and IOT Technologies, Engineering and R&D

- Led global delivery of medical device programs across Imaging, Diabetes care, Patient monitoring, IVD and connected healthcare platforms.
- Managed 250+ engineers, delivered multiple FDA 510(k) and ISO 13485-compliant products.
- Established medical device practice processes, including design history file management, regulatory documentation, and risk-based validation frameworks.
- Partnered with global clients (US, Japan, France) to shape product roadmaps, achieving multimillion-dollar growth through innovation-driven engineering.
- Build and lead team of systems, electrical, mechanical and firmware engineers
- Established regulatory Centers of Excellence and standardized compliance workflow, resulted in multiple EU-MDR remediation engagements and scaling the regulatory team to 75 professionals within 1 year.
- Designed procurement frameworks for COTS components critical for medical device development
- Orchestrated global engineering collaboration and knowledge-sharing practices, thus enhancing cross-site productivity and reduced resource bottlenecks.

Success stories:

1. New product development of a blood apheresis device - Unique features of this device besides performance are that it is a connected medical device and slated to be launched as platform as a service. Team owned electrical, mechanical and firmware development of this device.
2. Reliability lab set up - A state-of-the-art reliability characterization lab has been setup to find out early life cycle defects in electromechanical components during design phase. It helped find numerous critical defects that would have been undetected otherwise. Value add from this lab has been appreciated multiple times by our customer.
3. Value engineering – CT gantry control system

Cyient, Hyderabad India (May 2017 to Dec 2017)

Head Embedded software electronics

Manage P&L, delivery operations of ESE group. Key role and responsibilities include managing electronics product developments, customer engagement, managing delivery operation, profitability and talent management.

ODC (fortune 500 client) – provides firmware, electrical, mechanical design services across different business units of this customer. Projects include NPI and LCM. IIOT device was developed for industrial freezer device as part of NPI.

eInfoChips (an Arrow Company), Pune India (Jan 2016 to Apr 2017)

BU Head, Hardware design

eInfochips, an Arrow company, is a leading global product engineering and semiconductor design services provider. I managed P&L and delivery operations of hardware design business unit. Focus areas are consumer electronics, medical electronics and industrial automation. Our team has state of the art hardware IPs with Android, Linux BSPs which acts as a solution accelerator to achieve faster time to market. These IPs are driving faster time to market in new product designs.

Highlights:

1. Reusable auto injector design – capable of injecting 100CP drug in subcutaneous layer
2. Smart city solution – WiFi/4G enabled next generation light sensory network platform
3. Rugged intrinsically safe Qualcomm 410 based android mobile phone design
4. Managed Qualcomm SOC based System on module designs

IGATE/Capgemini, Pune, India (Jan 2005 to Jan 2016)

Associate Director (Delivery head, hardware design practice)

Managed delivery operations of hardware design group that addresses embedded product designs across industry verticals worldwide. The key role responsibilities include managing delivery operations, customer engagement, business development support, ensuring quality of delivery, on time within budget and talent management. Many offshore engineering centers were established for medical, industrial and automotive customers during this stint.

Significant Accomplishments

- Setup and nurtured hardware/embedded design practice in IGATE – 200+ people team
- Lead cross functional delivery team to support implementation of ISO 13485, ISO14971, ISO 62366 and end to end product development methodologies.
- Played a key role in setting up of reliability COE
- Institutionalize value engineering service offering
- Setup a systems engineering team to strengthen end to end service offering

Innovations

- Disposables tracking concept - helps authenticate and tracking of disposables used on medical devices
- Scalable test environment – facilitates complete design verification of hardware PCBs using LabVIEW framework. Reduced verification cycle time by 60%
- Usage of diagnostics imaging technique for rapid verification of invasive devices on phantoms
- Home grown component management system based on OmniFY PLM and Partminer/CAPS Expert

Key customers/products/services offered:

- Industrial automation major – MPC, VxWorks based mini PLC design
- Top blood management company – ODC caters to sustenance of all product lines and NPD
- Tier I automotive vendor – magic folding ECU design, reliability analysis
- Design for emerging markets – complete design of industry first visual stethoscope
- Medical imaging leader Japan – value engineering, reengineering
- NPD – continuous glucose monitoring design using Nordic SOC, Apple iOS app
- Engineering library management – team managing library of electronic components for global R&D centers of a customer
- NPD – bed side monitor design around Kinetis, uC3 RTOS and PEG UI
- Value engineering – reliability analysis and value engineering leading to savings of \$1.5MM per year, value engineering of high acuity ventilator
- NPD – power quality analyzer design using ARM, SHARC DSP, Linux, QT

Honeywell Automation India Ltd., Pune (Nov'02 to Jan'05)

Project Manager, Industrial Automation R&D

(Includes - Nov'02 to Jun'03 Transtech Ltd., Pune as Tech Lead at Honeywell PTS Group)

- Instrumental in acquiring a new customer in condition monitoring space

Setup a team to support PDA based vibration analyzer product line for this customer. Support included product enhancements and porting on new embedded visual studio, pocket PC, WinCE based PDA platforms.

- Head building automation India R&D team

We developed HVAC building automation controller for India market. Aim was to develop a low cost, low capacity HVAC controller which can work in building automation network along with existing devices.

Tata Infotech Ltd (Mar'01 – Nov'02) – Sr Systems Engineer

AE Telelink Systems Ltd, New Delhi (Mar'00 to Feb'01) – Project Manager

Maharashtra Electronics Corporation Ltd, Nagpur (Jul'89 to Feb'00) – R&D Engineer

Education

2015 - Indian Institute of Management, Lucknow – Executive General Management Program

1995 -Institute of Electronics and Telecom Engineers, New Delhi – BE E&TC

LinkedIn Profile: <http://www.linkedin.com/in/ravindra-todmal>

Trainings and Certifications

- ISO 13485 Implementation
- Agile (SCRUM, SAFe)
- RTOS based designs
- ASQ – reliability engineering
- EMI/EMC